IHDEA

2025 MEETING

CDPP / CNES HIGHLIGHTS

Danièle Boucon

HISTORY AND CONTEXT (REMINDER)

CDPP* activities are "historically" split into two connected activities

- Long Term preservation, under CNES responsibility
- Scientific valorization, under CNRS/INSU responsibility

➢ 2 separated databases

- The CDPP archive in CNES
- The AMDA tool database in IRAP

➢ 2 different data models “historically” created

- The "CDPP model" for the archive
- The "AMDA model"

➢ Limited consistency → Now both models are SPASE compliant. Still limited interfaces and possibilities for sophisticated interactions

** More information on CDPP server https://www.cdpp.eu/*

GOALS

Need to

- Standardization and coherence of CDPP data models
  - Be compliant to community standards
  - Use the same model for all the CDPP tools

- Coordinated actions between CNES and IRAP to achieve:
  - Better usability
  - Better interaction/integration between CDPP tools
  - Offer new possibilities, such as

  Using archive data directly in CDPP scientific tools

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Using archive data directly in CDPP scientific tools

ARCHIVE MAIN ACTIVITIES SINCE LAST IHDEA MEETING 1/3

- REGARDS new archiving system now operational
  - the data catalog proposes SPASE compliant metadata (22 missions + 2 new ones)
  - Data model conforms to SPASE 2.4.0 (compliant with 2.7)
  - the HMI has continued to evolve (since 2025)
  - better data search and retrieval performance
  - Link to access the CDPP archive (same as for SIPAD) : https://cdpp-archive.cnes.fr

If you have some problems, support address (new): cdpp@cnes.fr

REGARDS

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CDPP
CNES Data Archive for CDPP
General presentation
Missions and Products
Documents and Software
About
The CDPP is the french plasma physics data center. Created in 1998 by CNES and CNRS.
It assures the long term preservation of data obtained primarily from instruments built using French resources, and renders them readily accessible and exploitable by the international community.
The CNRS is responsible for the scientific component which is hosted at IRAP with the collaboration of Observatoire de Paris.
This site represents the archive of data hosted at CNES, which is responsible for their long-term preservation.
For more information please contact Support.
Data archive
Data access
Data available here are public but to order data you need to be registered (see user guide).
Please acknowledge the CDPP and the PI of the data if you use it in a publication, ...for more details read the template.
Other links & News
CDPP CNES archive data is opened since 2023.
Useful links : the main site of CDPP. The latter provides users with a number of tools such as : AMDA, 3DView, Propagation tool, spaceweather tool and Treps.
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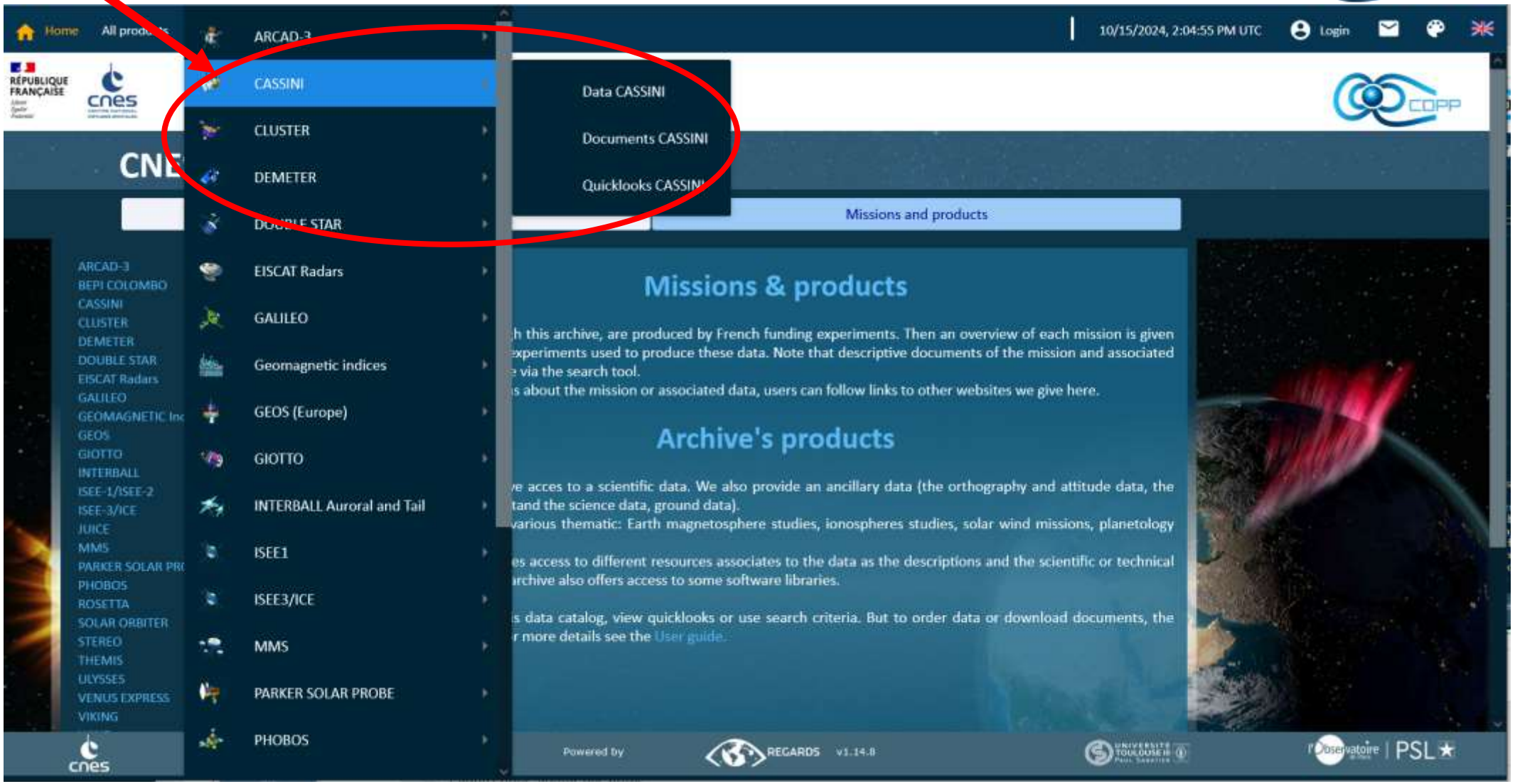
REGARDS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DOUBLE STAR
EISCAT Radars
GALILEO
Geomagnetic indices
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GIOTTO
INTERBALL Auroral and Tail
ISEE1
ISEE3/ICE
MMS
PARKER SOLAR PROBE
PHOBOS
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Documents CASSINI
Quicklooks CASSINI
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Data CASSINI | Results | Description

CAS_RPWS_HFR_QTN_N3_CDF | SEARCH RELATED DATA

Parameters

Information

- CASSINI Spase Descripti...
- Dataset Description
- Dataset Spase Descripti...
- Experiment Description
- Mission Description
- RPWS-HFR Instrument ...
- RPWS-HFR Instrument S...

CDPP.io

Data Access
- REGARDS

CASSINI RPWS L3 Electron Density and Temperature from Quasi-Thermal Noise Analysis on RPWS/HFR Spectra

ResourceID
spase://CNES/NumericalData/CDPP-Archive/CASSINI/RPWS-HFR/CAS_RPWS_HFR_QTN_N3_CDF

Description
This dataset contains electron density and temperature from quasi-thermal noise analysis on RPWS/HFR spectra.

Details View XML | Edit

Version:2.4.0

NumericalData

ResourceID
spase://CNES/NumericalData/CDPP-Archive/CASSINI/RPWS-HFR/CAS_RPWS_HFR_QTN_N3_CDF

ResourceHeader

ResourceName
CASSINI RPWS L3 Electron Density and Temperature from Quasi-Thermal Noise Analysis on RPWS/HFR Spectra

DOI
https://doi.org/10.24400/802406/vqt7-1k9p

ReleaseDate
2022-06-02 17:57:30Z

Description
This dataset contains electron density and temperature from quasi-thermal noise analysis on RPWS/HFR spectra.

Acknowledgement
Data provided are publicly available. If used in presentations or publications, please acknowledge Cassini/RPWS investigation (W. S. Kurth (PI) Univ. of Iowa).
We recommend the data users to contact the Cassini/RPWS team early in their study. For reporting purposes, we request bibliography information for any publication, etc., using these data. Please send information on the use of these data to:

PI : William S. Kurth (william-kurth@uiowa.edu)
co-I : Baptiste Cecconi (baptiste.cecconi@obspm.fr)

We suggest to add the following lines into the acknowledgments:

The authors acknowledge NASA, University of Iowa, CNES (Centre National d'Etudes Spatiales), CNRS (Centre National de la Recherche Scientifique), and Observatoire de Paris for their support to the Cassini/RPWS team and the CDPP (Centre de Données de la Physique des Plasmas) for the distribution of the Cassini/RPWS data.

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CDPP.io

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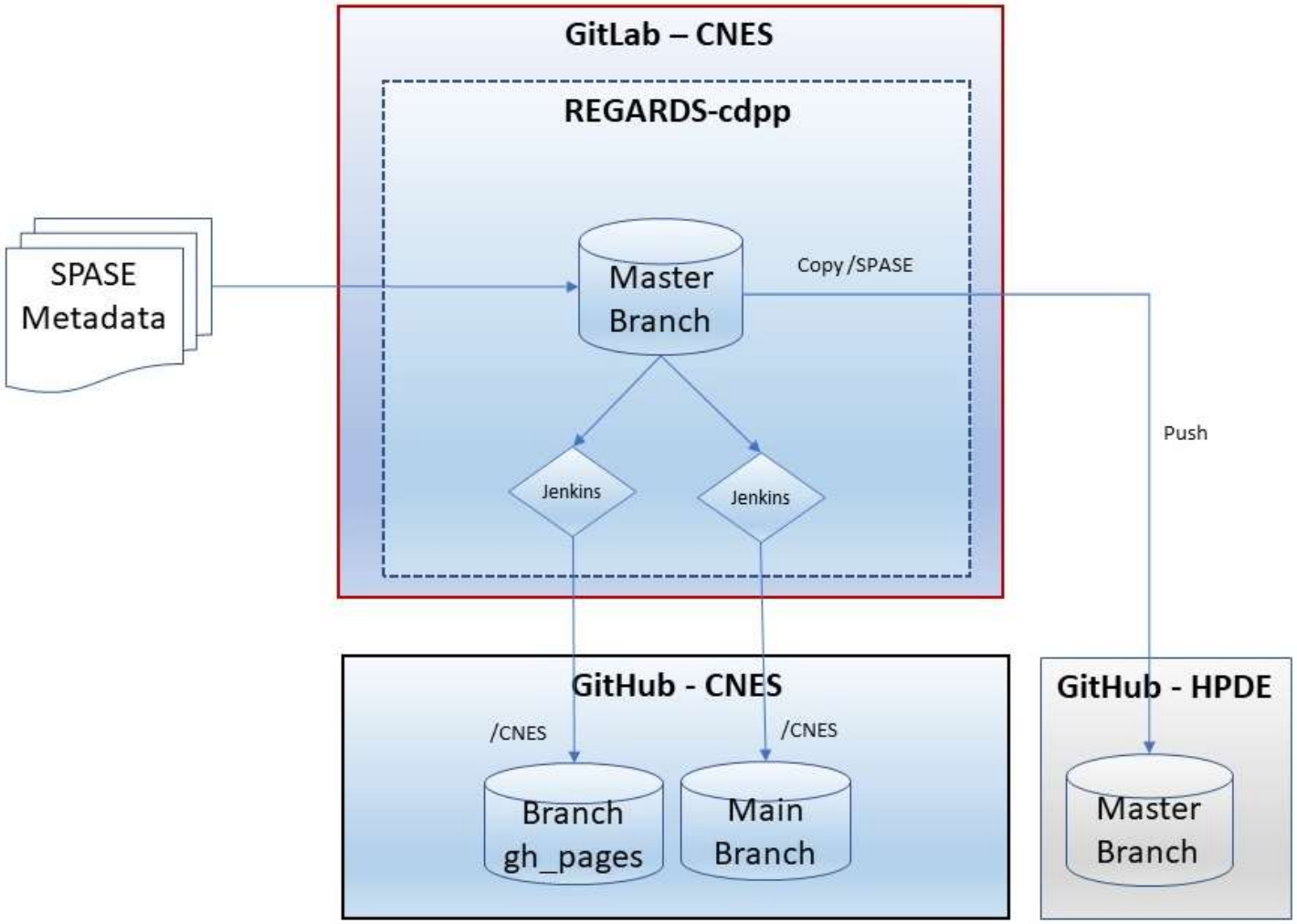
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Master Branch
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GitHub - CNES
/CNES
/CNES
Branch gh_pages
Main Branch
GitHub - HPDE
Master Branch

ARCHIVE MAIN ACTIVITIES SINCE LAST IHDEA MEETING 2/3

- SPASE landing pages repository https://spase-metadata.org
  - All CNES SPASE metadata on this repository
  - Direct access to the data through the landing page

Ex CASSINI / RPWS HFR

https://spase-metadata.org/CNES/NumericalData/CDPP-Archive/CASSINI/RPWS-HFR/CAS_RPWS_HFR_QTN_N3_CDF.html

Direct data access

spase-metadata.org SPASE landing pages.

Data Access

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ResourceName
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Contacts

Role	Person	StartDate	StopDate	Note
1. PrincipalInvestigator	spase://CNES/Person/CDPP-Archive/William.S.Kurth			
2. DeputyPI	spase://CNES/Person/CDPP-Archive/Donald.A.Gurnett			
3. CoInvestigator	spase://CNES/Person/CDPP-Archive/Baptiste.Cecconi			
4. CoInvestigator	spase://CNES/Person/CDPP-Archive/Philippe.M.Zarka			
5. CoInvestigator	spase://CNES/Person/CDPP-Archive/Laurent.Lamy			

AccessInformation

RepositoryID
spase://CNES/Repository/CDPP-Archive/REGARDS-CDPP

Availability
Online

AccessURL

Name
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URL
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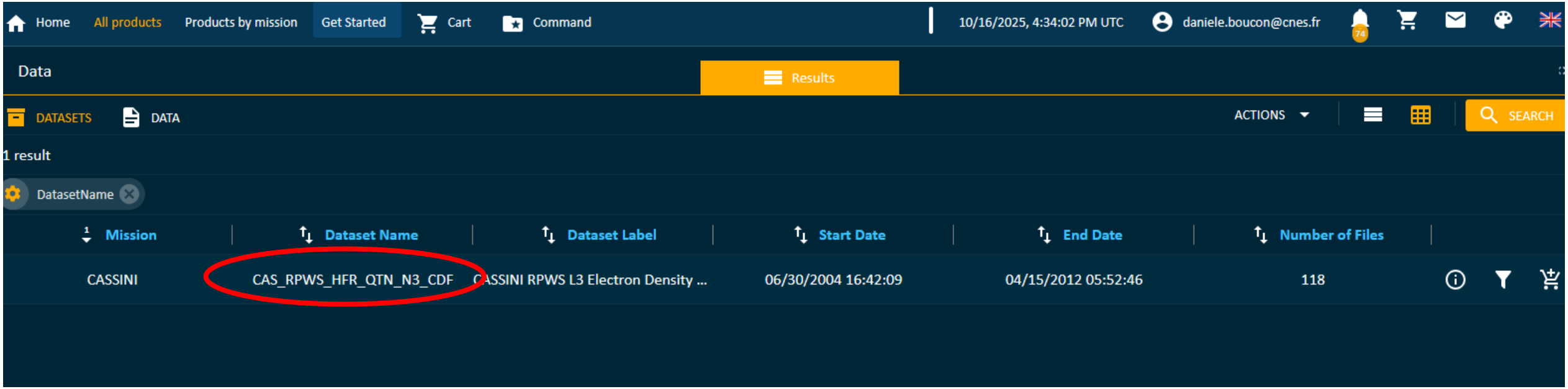
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Start Date
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ARCHIVE MAIN ACTIVITIES SINCE LAST IHDEA MEETING 3/3

- **MAVEN/SWEA and SOLAR ORBITER / RPW**

Work with the scientist to harmonize the NumericalData descriptions (ESA and CNES/Solar O.)

ARCHIVE ROADMAP

➢1st phase (from now)

- **Moving to a new long term storage system**: old data analysis, data migration -> 2027
- **Integration of new missions (short term MAVEN and Solar Orbiter, mid and long term JUICE and BEPI COLOMBO)**

➢2nd phase (from 2026)

- REGARDS will continue to evolve (scientific community feedback…)
- **Integration with CDPP valorization tools will be reinforced**

  - **Archive data in the new storage infrastructure will be accessible by the CDPP tools (webservices, VO protocols)**
  - **Visualization and retrieval of the AMDA data through the Archive**
    Common search with common criteria (ex. search all Cassini data in both databases)

ARCHIVE ROADMAP

2025: old data analysis, data migration

Integration of new missions

2027:

new long term storage system

From 2026:

REGARDS + CDPP valorization tools reinforced

https://cdpp-archive.cnes.fr

THANK YOU FOR YOUR ATTENTION